

Notes on the Oracle-Averaging Conjecture

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Abstract

Let O be a random oracle with independent $\{\pm 1\}$ outputs, and let f be a deterministic oracle algorithm that, on input $x \in \{0, 1\}^N$, makes at most k oracle queries and outputs a value in $[-1, 1]$. Writing

$$F(O) := \bar{f}(O) = 2^{-N} \sum_{x \in \{0, 1\}^N} f^O(x),$$

and letting E_d be the minimum mean-squared error achievable by an estimator with at most dk total oracle queries, the conjecture is that $\sum_{d \geq 1} E_d \leq 1$. This note records three points: (i) the conjecture is true in the oblivious (non-adaptive query-set) case; (ii) in full generality one always has the weaker bound $\sum_{d \geq 1} E_d \leq k$; and (iii) for a layered test family, a merge-aware budgeted estimator appears empirically consistent with the conjectured 1-bound.

1 Problem statement

Let $\mathcal{Q}$ be a countable query set and let

$$O : \mathcal{Q} \rightarrow \{-1, +1\}$$

be a random oracle, with the values $\{O(q) : q \in \mathcal{Q}\}$ independent and uniform in $\{-1, +1\}$. Throughout, f is a *deterministic* oracle algorithm. On input $x \in \{0, 1\}^N$, it makes at most k oracle queries and outputs a value $f^O(x) \in [-1, 1]$.

Define

$$F(O) = \bar{f}(O) := \frac{1}{2^N} \sum_{x \in \{0, 1\}^N} f^O(x).$$

For each integer $d \geq 1$, let E_d be the Bayes risk

$$E_d := \inf_{\hat{\mu}} \mathbb{E}[(\hat{\mu} - F(O))^2],$$

where the infimum runs over all (possibly randomized, adaptive) estimators $\hat{\mu}$ that make at most dk oracle queries in total. The expectation is over the oracle and the estimator's internal randomness.

The conjecture under discussion is:

$$\sum_{d=1}^{\infty} E_d \leq 1.$$

A trivial bound is $E_d \leq 1/d$: sample d independent uniformly random inputs $X_1, \dots, X_d$, run f on each, and average the outputs. This uses at most dk oracle queries.

The sections below give two rigorous positive results and then discuss a structured test family.

2 The oblivious case

We say that f is *oblivious* if for each input x there is a fixed set $S_x \subseteq \mathcal{Q}$ with $|S_x| \leq k$ and a function

$$h_x : \{-1, +1\}^{S_x} \rightarrow [-1, 1]$$

such that

$$f^O(x) = h_x((O(q))_{q \in S_x}).$$

Thus the queried locations may depend on x , but not on the oracle answers.

Theorem 1. *If f is oblivious, then*

$$\sum_{d=1}^{\infty} E_d \leq 1.$$

In fact,

$$\sum_{d=1}^{\infty} E_d \leq \frac{1}{2^N} \sum_x \text{Var}(f^O(x)) \leq 1.$$

Proof. Let

$$Q := \bigcup_x S_x,$$

which is finite. For each $T \subseteq Q$, define the Walsh character

$$\chi_T(O) := \prod_{q \in T} O(q).$$

Each h_x has an expansion

$$h_x = \sum_{T \subseteq S_x} \hat{h}_x(T) \chi_T.$$

Hence

$$F = \sum_{T \subseteq Q} a_T \chi_T, \quad a_T := 2^{-N} \sum_x \hat{h}_x(T).$$

Fix $d \geq 1$. Sample $X_1, \dots, X_d$ independently and uniformly from $\{0, 1\}^N$, and let

$$U_d := S_{X_1} \cup \dots \cup S_{X_d}.$$

Query all oracle values in U_d and output the posterior mean

$$\hat{\mu}_d := \mathbb{E}[F \mid X_1, \dots, X_d, O|_{U_d}].$$

This estimator uses at most dk oracle queries, so its risk upper-bounds E_d .

For fixed $U \subseteq Q$,

$$\mathbb{E}[F \mid O|_U] = \sum_{T \subseteq U} a_T \chi_T,$$

because $\mathbb{E}[\chi_T \mid O|_U] = 0$ unless $T \subseteq U$. Therefore

$$F - \mathbb{E}[F \mid O|_U] = \sum_{T \not\subseteq U} a_T \chi_T,$$

and orthogonality of the Walsh basis gives

$$\mathbb{E}\left[(F - \mathbb{E}[F \mid O|_U])^2\right] = \sum_{T \notin U} a_T^2.$$

Averaging over the random set U_d ,

$$E_d \leq \sum_{T \neq \emptyset} a_T^2 \mathbb{P}[T \notin U_d].$$

Now define

$$p_T := \mathbb{P}_{X \sim \text{Unif}}[T \subseteq S_X].$$

If $p_T = 0$ then $\hat{h}_x(T) = 0$ for every x , hence $a_T = 0$. So only $p_T > 0$ matters. If some sampled input contains T , then certainly $T \subseteq U_d$, and therefore

$$\mathbb{P}[T \notin U_d] \leq (1 - p_T)^d.$$

Summing in d yields

$$\sum_{d=1}^{\infty} E_d \leq \sum_{T \neq \emptyset} a_T^2 \sum_{d=1}^{\infty} (1 - p_T)^d \leq \sum_{T \neq \emptyset} \frac{a_T^2}{p_T}.$$

Finally,

$$a_T = \mathbb{E}_X[\mathbf{1}_{\{T \subseteq S_X\}} \hat{h}_X(T)],$$

so Cauchy-Schwarz gives

$$a_T^2 \leq p_T \mathbb{E}_X[\mathbf{1}_{\{T \subseteq S_X\}} \hat{h}_X(T)^2].$$

Hence

$$\frac{a_T^2}{p_T} \leq \mathbb{E}_X[\mathbf{1}_{\{T \subseteq S_X\}} \hat{h}_X(T)^2].$$

Summing over $T \neq \emptyset$ and using Parseval,

$$\sum_{d=1}^{\infty} E_d \leq \mathbb{E}_X \sum_{T \subseteq S_X, T \neq \emptyset} \hat{h}_X(T)^2 = \mathbb{E}_X \text{Var}(h_X(O)) = \frac{1}{2^N} \sum_x \text{Var}(f^O(x)) \leq 1.$$

□

3 A general bound of k

The oblivious proof keeps the entire Fourier support of each input together. In the adaptive case that structure is lost, but one can still prove a clean worst-case bound.

Theorem 2. *For every deterministic depth- k oracle algorithm f ,*

$$\sum_{d=1}^{\infty} E_d \leq k.$$

Proof. For each stage $d \geq 1$, sample $X_1, \dots, X_d$ independently and uniformly from $\{0, 1\}^N$, run f on each sampled input, record the full transcript T_d , and output the posterior mean

$$\hat{\mu}_d := \mathbb{E}[F \mid T_d].$$

This uses at most dk oracle queries, so

$$E_d \leq \mathbb{E}[(\hat{\mu}_d - F)^2].$$

It remains to bound the sum of these risks.

Only finitely many oracle points are relevant, because there are finitely many inputs and each input gives a depth- k decision tree. Fix one such point q . Define

$$A_{x,q}(O) := \mathbf{1}_{\{\text{the run of } f^{O(x)} \text{ queries } q\}}, \quad \rho_q(O) := 2^{-N} \sum_x A_{x,q}(O).$$

Thus $\rho_q(O)$ is the fraction of inputs whose run touches q .

Since each input uses at most k queries,

$$\sum_q \rho_q(O) = 2^{-N} \sum_x \sum_q A_{x,q}(O) \leq 2^{-N} \sum_x k = k. \quad (1)$$

Next define the discrete derivative

$$D_q F(O) := \frac{F(O^{q \rightarrow +1}) - F(O^{q \rightarrow -1})}{2}.$$

For fixed x , the event “the run on input x queries q ” is independent of the value $O(q)$: before the first possible query to q , the transcript depends only on other oracle values. Hence either both runs $O^{q \rightarrow +1}$ and $O^{q \rightarrow -1}$ query q , or neither does. Therefore

$$|D_q f^O(x)| \leq A_{x,q}(O),$$

and averaging over x gives

$$|D_q F(O)| \leq \rho_q(O). \quad (2)$$

Let R_d be the set of oracle points queried in the transcript T_d . Conditioned on T_d , the queried oracle values are fixed and the unqueried values remain independent fair signs. Applying the Poincaré/Efron-Stein inequality on the remaining coordinates,

$$\text{Var}(F \mid T_d) \leq \sum_{q \notin R_d} \mathbb{E}[(D_q F)^2 \mid T_d].$$

Taking expectation and using the posterior mean identity,

$$\mathbb{E}[(\hat{\mu}_d - F)^2] = \mathbb{E}[\text{Var}(F \mid T_d)] \leq \sum_q \mathbb{E}[(D_q F)^2 \mathbf{1}_{\{q \notin R_d\}}]. \quad (3)$$

Now condition on the oracle O . The indicators of whether q is touched by the d sampled inputs are i.i.d. Bernoulli with mean $\rho_q(O)$, so

$$\mathbb{P}(q \notin R_d \mid O) = (1 - \rho_q(O))^d.$$

Substituting into (3),

$$\mathbb{E}[(\hat{\mu}_d - F)^2] \leq \sum_q \mathbb{E}[(D_q F)^2 (1 - \rho_q)^d].$$

Summing over $d \geq 1$ and using the geometric series,

$$\sum_{d=1}^{\infty} \mathbb{E}[(\hat{\mu}_d - F)^2] \leq \sum_q \mathbb{E}\left[(D_q F)^2 \frac{1 - \rho_q}{\rho_q}\right],$$

with the convention that the term is 0 when $\rho_q = 0$ (by (2), then also $D_q F = 0$).

Finally, (2) implies $(D_q F)^2 \leq \rho_q^2$, so

$$(D_q F)^2 \frac{1 - \rho_q}{\rho_q} \leq \rho_q (1 - \rho_q) \leq \rho_q.$$

Therefore

$$\sum_{d=1}^{\infty} E_d \leq \sum_{d=1}^{\infty} \mathbb{E}[(\hat{\mu}_d - F)^2] \leq \sum_q \mathbb{E}[\rho_q] \leq k$$

by (1). □

4 A layered test family

The following family is a useful stress test. It is not literally the original $\{\pm 1\}$ -oracle model, because intermediate oracle values lie in $[N]$ rather than $\{\pm 1\}$, but it can be encoded in the original setting with an extra $O(\log N)$ overhead per transition. The point here is empirical: it is a structured family in which merging trajectories creates a large amount of information reuse.

Fix a width N and depth L . For each layer $i = 0, \dots, L - 1$, let

$$T_i : [N] \rightarrow [N]$$

be an independent uniformly random map, and let

$$s : [N] \rightarrow \{-1, +1\}$$

be independent uniformly random terminal signs. For an input $x_0 \in [N]$, define a path by

$$x_{i+1} = T_i(x_i) \quad (0 \leq i < L)$$

and output

$$f(x_0) = s(x_L).$$

A fresh start uses $k = L + 1$ oracle lookups.

Merge-aware budgeted estimator

The estimator used in the experiments keeps a random order of the N start vertices. It explores paths one oracle lookup at a time, caching every revealed transition and sign. If the current path reaches a vertex whose entire remaining suffix is already known, the path is retired immediately and the estimator moves to the next start. After every budget increment of $(L + 1)$ oracle queries—that

is, after total budget $(L + 1)d$ —the estimator outputs the posterior mean of F given the current transcript.

The simulator computes the posterior mean and variance exactly from the partial transcript by propagating first and second moments backwards through the partially revealed layered graph. The only Monte Carlo is over the sampled oracle and the random start order.

There are two natural ways to index the resulting risks:

- **Budgeted-by-queries:** stage d means total budget $(L + 1)d$, which matches the conjecture’s resource accounting.
- **By completed starts:** stage d means that d starts have been retired, regardless of how many queries were actually used. This is a useful comparison baseline, but it is not the quantity appearing in E_d .

Empirical observations

Figure 1 shows a prefix-depth sweep for widths $N \in \{16, 32, 64, 128\}$. For each width, one depth-16 oracle was sampled per trial and then reused for all prefix depths $L' \leq 16$, so the same computation simultaneously yields the first L' layers for every L' . The merge-aware budgeted estimator stayed below 1 throughout this sweep.

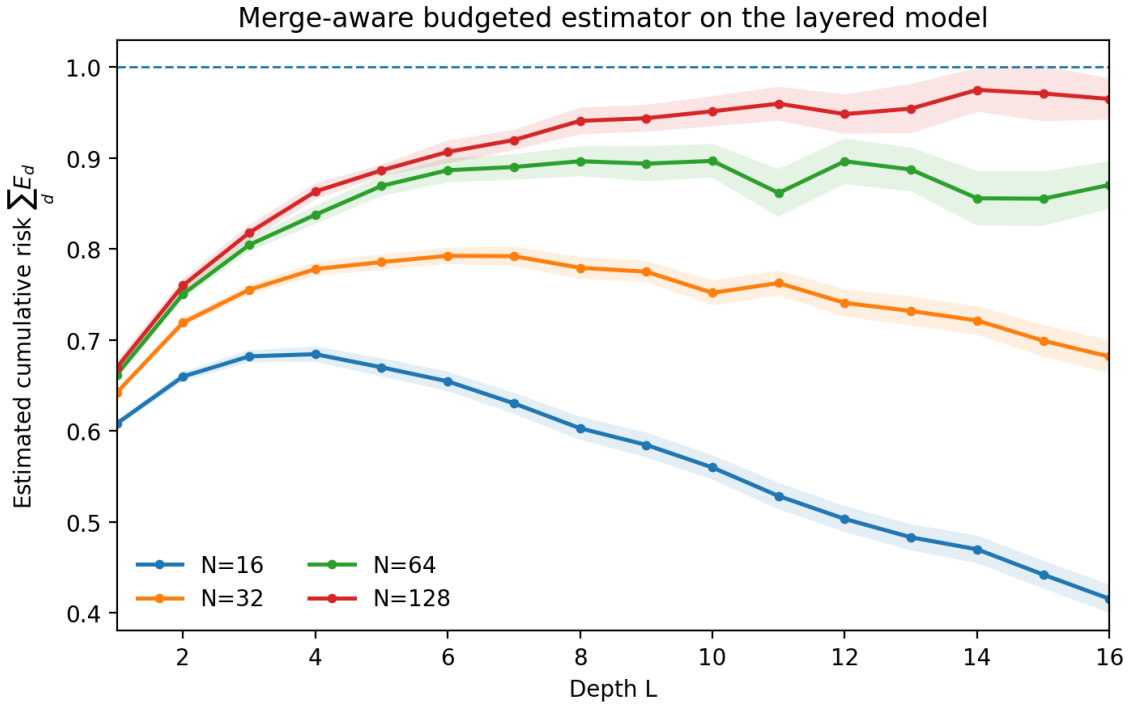


Figure 1: Estimated cumulative risk of the merge-aware budgeted estimator on the layered model. Shaded bands show empirical 95% confidence intervals over oracle samples.

The peak values in that sweep were:

Table 1: Peak empirical values in the depth-16 prefix sweep.

Width N	Repetitions	Peak depth L	Peak mean	95% CI
16	250	4	0.684	[0.676, 0.693]
32	160	6	0.792	[0.784, 0.801]
64	40	10	0.897	[0.878, 0.915]
128	10	14	0.975	[0.951, 0.999]

A more focused run at $(L, N) = (32, 128)$, averaged over 20 oracle samples, gave:

- budgeted-by-queries cumulative mean 0.9598 ± 0.0176 (standard error), with empirical 95% interval $[0.9252, 0.9943]$;
- completed-starts cumulative mean 1.8645 ± 0.0355 ;
- average total number of unique queries needed to finish all 128 starts: 764.2 ± 7.0 , compared with the no-reuse cost $128 \cdot 33 = 4224$.

Figure 2 compares the two indexings. The completed-start curve is far above 1, while the budgeted-by-queries curve stays below 1 on average.

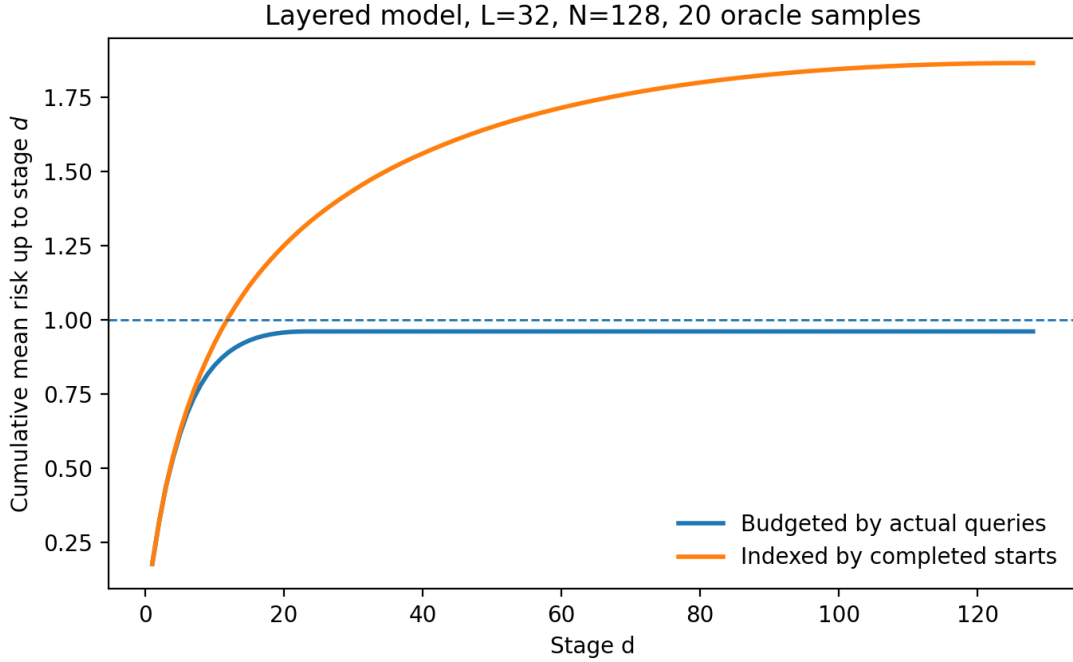


Figure 2: Layered model at $L = 32$, $N = 128$, averaged over 20 oracle samples. The conjecture concerns the blue curve, which is indexed by actual query budget.

The same run is still not a proof. Individual oracle samples can lie above 1 even when the empirical mean is below it: in this experiment 7 of the 20 sampled oracles had budgeted cumulative sum greater than 1, with sample range $[0.793, 1.098]$. Since the conjecture is about expectation over the oracle, the average is the relevant quantity, but the variance here is not negligible.

5 Current status

The picture at this point is:

- The conjectured bound $\sum_{d \geq 1} E_d \leq 1$ is proved for oblivious machines.
- In full generality, the weaker but unconditional bound $\sum_{d \geq 1} E_d \leq k$ holds.
- For the layered family, the naive “count completed starts” view fails badly, but a merge-aware estimator indexed by actual oracle-query budget is empirically consistent with the 1-bound on the experiments run so far.

The remaining gap is therefore qualitative rather than computational: the simulations suggest that the right resource accounting really does have a chance to satisfy the conjectured unit bound, but a proof beyond the oblivious case is still missing.

Data and code used for the experiments. The simulation script and the raw summary tables are included alongside this note. The main files are:

- `layered_merge_simulation.py`
- `layered_merge_prefix_grid_combined.csv`
- `l32_n128_20oracle_summary.csv`
- `l32_n128_20oracle_per_sample.csv`
- `l32_n128_20oracle_stagewise.csv`